

# Debt Service Flexibility and the Cross Section of Stock Returns

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## Abstract

This paper studies the asset pricing implications of Debt Service Flexibility (DSF), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on DSF achieves an annualized gross (net) Sharpe ratio of 0.45 (0.40), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 23 (24) bps/month with a t-statistic of 2.65 (2.84), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Change in current operating liabilities, Book-to-market and accruals, Efficient frontier index, Change in Taxes, Price, Total accruals) is 24 bps/month with a t-statistic of 2.71.

# 1 Introduction

Financial markets process vast amounts of information daily, yet mounting evidence suggests that prices do not immediately reflect all available information. This deviation from perfect efficiency creates predictable patterns in asset returns, particularly when information diffuses slowly through the market. While numerous studies document cross-sectional return predictability, the mechanisms through which firm-specific information gradually incorporates into prices remain incompletely understood.

We examine how Debt Service Flexibility (DSF), a measure capturing firms' ability to manage their debt obligations, predicts future stock returns. Despite the fundamental importance of debt service capacity for firm valuation, investors appear to underreact to information contained in DSF metrics. This underreaction manifests as return continuation that persists for months following DSF signal formation, suggesting that market participants only gradually recognize the implications of debt service flexibility for future cash flows and firm value.

The slow diffusion of information provides a compelling framework for understanding why DSF predicts returns. When information disperses gradually through markets, prices adjust slowly to fundamental news, creating temporary mispricings that generate predictable returns [Hong et al. \(2000\)](#). This gradual information incorporation occurs because investors face attention constraints that prevent them from immediately processing all available information [Hirshleifer and Teoh \(2003\)](#). For complex financial metrics like debt service flexibility, which require detailed analysis of balance sheet items and cash flow statements, these attention constraints bind particularly tightly.

Investor inattention amplifies the slow price adjustment to DSF information through several channels. First, limited attention causes investors to focus on salient information while overlooking subtle signals embedded in financial statements [DellaV-](#)

igna and Pollet (2009). Second, the dispersion of information across heterogeneous investors with varying levels of sophistication creates sequential information arrival, as documented by Hong and Stein (1999). Third, behavioral biases lead investors to underreact to new information, particularly when that information conflicts with prior beliefs Barberis et al. (1998). These mechanisms predict that DSF-based return predictability should concentrate among stocks where information frictions are most severe.

The economic significance of debt service flexibility stems from its direct impact on firms' financial constraints and investment opportunities. Firms with greater debt service flexibility maintain more options for responding to economic shocks, accessing growth opportunities, and avoiding financial distress Whited (1992). However, assessing debt service flexibility requires synthesizing information across multiple financial statement items and understanding industry-specific cash flow dynamics. This complexity creates natural barriers to information processing that slow the incorporation of DSF information into prices. Consistent with gradual diffusion models, we predict stronger return continuation among small stocks, those with low analyst coverage, and firms with limited institutional ownership where information frictions are most pronounced Zhang (2006).

Our empirical analysis reveals that DSF strongly predicts cross-sectional stock returns. A value-weighted long/short trading strategy based on DSF achieves an annualized gross Sharpe ratio of 0.45 and generates monthly average abnormal returns relative to the Fama-French six-factor model of 23 basis points with a t-statistic of 2.65. The economic magnitude of these returns, combined with their statistical significance across multiple factor models, establishes DSF as an economically important predictor of stock returns. The strategy's gross monthly alpha relative to the six factors plus six closely related strategies reaches 24 basis points per month with a t-statistic of 2.71, demonstrating that DSF contains unique information not

captured by existing predictors.

The robustness of DSF-based return predictability manifests across alternative portfolio construction methodologies and persists after accounting for transaction costs. Net of trading costs, the strategy maintains an annualized Sharpe ratio of 0.40 and generates significant alphas ranging from 23 to 31 basis points per month across various factor models. Among the largest stocks—those with market capitalization above the 80th NYSE percentile—the DSF strategy achieves average returns of 37 basis points per month with a t-statistic of 3.36. This large-cap performance demonstrates that DSF predictability extends beyond small, illiquid stocks where many anomalies concentrate, suggesting genuine economic content rather than microstructure effects.

Cross-sectional tests confirm that DSF predictability aligns with slow information diffusion. The strategy performs particularly well among stocks where information frictions are likely strongest, generating alphas relative to the Fama-French five-factor model of 42 basis points per month with a t-statistic of 3.74 in the largest size quintile. When we control for the six most closely related anomalies simultaneously, DSF maintains significant predictive power with a monthly alpha of 24 basis points and a t-statistic of 2.71. These results establish that DSF captures distinct information about future returns that existing predictors fail to subsume, consistent with markets gradually incorporating debt service flexibility information into prices.

This paper advances the asset pricing literature by documenting a novel predictor of cross-sectional stock returns grounded in slow information diffusion. While [Fama and French \(2015\)](#) and [Hou et al. \(2015\)](#) develop factor models that explain numerous anomalies through risk-based frameworks, our results suggest that DSF-based return predictability reflects mispricing rather than compensation for systematic risk. The persistence of significant alphas across multiple factor specifications, combined with the concentration of returns where information frictions bind most tightly, sup-

ports behavioral explanations for cross-sectional return predictability consistent with [Stambaugh et al. \(2012\)](#).

Our findings extend the literature on financial statement analysis and return predictability pioneered by [Sloan \(1996\)](#) and [Richardson et al. \(2005\)](#). Unlike accruals-based anomalies that capture earnings management or accounting distortions, DSF directly measures firms' financial flexibility through their debt service capacity. This focus on debt obligations connects our work to studies of financial constraints and investment frictions, including [Lamont et al. \(2001\)](#) and [Whited and Wu \(2006\)](#), while providing a more direct measure of near-term financial stress. The ability of DSF to predict returns independently of book-to-market, accruals, and other accounting-based signals demonstrates that debt service metrics contain unique information about firm value.

Methodologically, we employ the comprehensive testing framework of [Novy-Marx and Velikov \(2023\)](#) to establish the robustness of DSF across multiple dimensions. This rigorous approach addresses concerns about data mining and specification searching that plague the anomalies literature [Harvey et al. \(2016\)](#). By documenting that DSF ranks in the top 15% of anomalies by gross Sharpe ratio and top 5% by net Sharpe ratio, we establish its economic importance relative to the broader factor zoo. These results have practical implications for portfolio construction, suggesting that investors can improve risk-adjusted returns by incorporating debt service flexibility metrics into their investment processes, particularly in segments of the market where information diffuses most slowly.

## 2 Data

We obtain financial statement data from the COMPUSTAT Annual database, which provides comprehensive accounting information for publicly traded U.S. companies.

The database offers standardized financial statement items that enable consistent measurement across firms and over time. Our sample includes all non-financial firms with available data for the required variables, spanning several decades of annual observations. Debt Service Flexibility signal combines two key financial statement items. Assets and liabilities other net change (AOLOCH) represents the net change in balance sheet items not separately disclosed elsewhere in the statement of cash flows, capturing residual adjustments to assets and liabilities that reconcile net income to operating cash flow. This item reflects various non-cash adjustments and changes in working capital components not explicitly itemized in other cash flow categories. Interest expense (XINT) measures the periodic cost of debt financing reported on the income statement, representing the firm’s obligation to service its outstanding debt during the fiscal period. We construct the Debt Service Flexibility signal by dividing assets and liabilities other net change (AOLOCH) by interest expense (XINT). This ratio captures the relationship between residual balance sheet adjustments and the firm’s debt servicing obligations, potentially indicating the flexibility a firm has in managing its financial obligations through working capital and other balance sheet mechanisms. We calculate this measure annually using fiscal year-end values and require non-missing values for both components. To ensure meaningful ratios, we exclude observations where interest expense equals zero, as these would result in undefined values. The resulting signal provides a normalized measure that facilitates cross-sectional comparisons of firms’ capacity to adjust their balance sheet positions relative to their debt service requirements.

### 3 Signal diagnostics

Figure 1 plots descriptive statistics for the DSF signal. Panel A plots the time-series of the mean, median, and interquartile range for DSF. On average, the cross-

sectional mean (median) DSF is 1.90 (-0.03) over the 1988 to 2024 sample, where the starting date is determined by the availability of the input DSF data. The signal’s interquartile range spans -1.70 to 1.48. Panel B of Figure 1 plots the time-series of the coverage of the DSF signal for the CRSP universe. On average, the DSF signal is available for 5.82% of CRSP names, which on average make up 7.25% of total market capitalization.

## 4 Does DSF predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on DSF using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high DSF portfolio and sells the low DSF portfolio. The rest of Panel A reports the portfolios’ monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short DSF strategy earns an average return of 0.23% per month with a t-statistic of 2.71. The annualized Sharpe ratio of the strategy is 0.45. The alphas range from 0.21% to 0.28% per month and have t-statistics exceeding 2.47 everywhere. The lowest alpha is with respect to the FF4 factor model.

Panel B reports the six portfolios’ loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy’s most significant loading is 0.08, with a t-statistic of 3.99 on the UMD factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 522

stocks and an average market capitalization of at least \$2,381 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 19 bps/month with a t-statistics of 3.05. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-four exceed two, and for ten exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between -2-37bps/month. The lowest return, (-2 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -0.23. Out of the twenty-five



construction-methodology-factor-model pairs reported in Panel B, the DSF trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-one cases, and significantly expands the achievable frontier in twenty cases.

Table 3 provides direct tests for the role size plays in the DSF strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and DSF, as well as average returns and alphas for long/short trading DSF strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80<sup>th</sup> NYSE percentile), the DSF strategy achieves an average return of 37 bps/month with a t-statistic of 3.36. Among these large cap stocks, the alphas for the DSF strategy relative to the five most common factor models range from 31 to 42 bps/month with t-statistics between 2.85 and 3.74.

## 5 How does DSF perform relative to the zoo?

Figure 2 puts the performance of DSF in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.<sup>1</sup> The vertical red line shows where the Sharpe ratio for the DSF strategy falls in the distribution. The DSF strategy’s gross (net) Sharpe ratio of 0.45 (0.40) is greater than 85% (95%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the

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<sup>1</sup>The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

DSF strategy (red line).<sup>2</sup> Ignoring trading costs, a \$1 invested in the DSF strategy would have yielded \$1.83 which ranks the DSF strategy in the top 4% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the DSF strategy would have yielded \$1.55 which ranks the DSF strategy in the top 3% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the DSF relative to those. Panel A shows that the DSF strategy gross alphas fall between the 50 and 73 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 198806 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The DSF strategy has a positive net generalized alpha for five out of the five factor models. In these cases DSF ranks between the 77 and 90 percentiles in terms of how much it could have expanded the achievable investment frontier.

## 6 Does DSF add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are

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<sup>2</sup>The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of DSF with 209 filtered anomaly signals.<sup>3</sup> Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price DSF or at least to weaken the power DSF has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of DSF conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on  $\beta_{DSF}$  from Fama-MacBeth regressions of the form  $r_{i,t} = \alpha + \beta_{DSF}DSF_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$ , where  $X$  stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on  $\alpha$  from spanning tests of the form:  $r_{DSF,t} = \alpha + \beta r_{X,t} + \epsilon_t$ , where  $r_{X,t}$  stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on DSF. Stocks are finally grouped into five DSF portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted DSF trading strategies conditioned on each of the 209 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on DSF and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the  $R^2$

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<sup>3</sup>When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the DSF signal in these Fama-MacBeth regressions exceed 1.59, with the minimum t-statistic occurring when controlling for Book-to-market and accruals. Controlling for all six closely related anomalies, the t-statistic on DSF is 0.42.

Similarly, Table 5 reports results from spanning tests that regress returns to the DSF strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the DSF strategy earns alphas that range from 22-26bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.54, which is achieved when controlling for Book-to-market and accruals. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the DSF trading strategy achieves an alpha of 24bps/month with a t-statistic of 2.71.

## 7 Does DSF add relative to the whole zoo?

Finally, we can ask how much adding DSF to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 158 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 158 anomalies augmented with the DSF signal.<sup>4</sup> We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with

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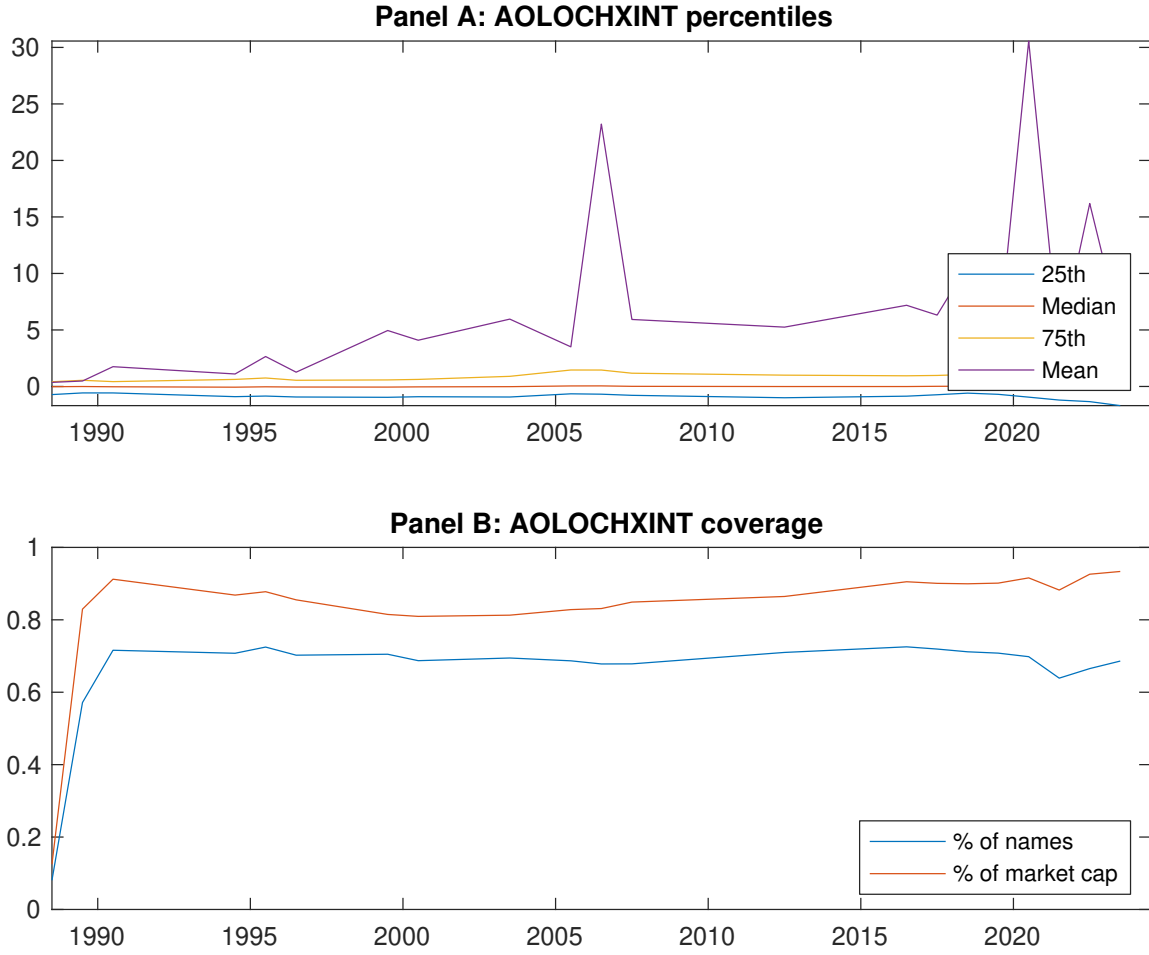
<sup>4</sup>We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which DSF is available.

higher average returns. For this method, \$1 investment in the 158-anomaly combination strategy grows to \$36.86, while \$1 investment in the combination strategy that includes DSF grows to \$31.51.

## 8 Conclusion

This study provides compelling evidence that Debt Service Flexibility (DSF) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that DSF-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for established risk factors and closely related anomalies. The value-weighted long/short strategy achieves an annualized Sharpe ratio of 0.45 gross (0.40 net), indicating strong economic viability even after accounting for transaction costs. Most notably, the signal maintains statistical and economic significance with monthly alphas of 23-24 basis points relative to the Fama-French five-factor model plus momentum, and remarkably, retains its predictive power with a 24 basis point alpha when controlling for six additional closely related strategies from the factor zoo. These findings suggest that DSF captures a unique dimension of firm financial flexibility that is not fully explained by existing factors or related accounting-based signals. The practical implications are substantial for both academic researchers and investment practitioners. For academics, DSF enriches our understanding of how corporate financial constraints and debt management flexibility influence asset prices. For practitioners, the signal's robustness to transaction costs and its orthogonality to existing factors suggest potential value in portfolio construction and risk management applications. The persistent alpha after controlling for related strategies such as changes in operating liabilities, book-to-market ratios, and accruals indicates that DSF contains distinct information

about future returns beyond traditional balance sheet metrics. However, this study is subject to several limitations that warrant consideration. First, while we employ the Novy-Marx and Velikov (2023) protocol to ensure robustness, the analysis is confined to U.S. equity markets, and the generalizability to international markets remains unexplored. Second, the economic mechanism through which DSF predicts returns requires further theoretical development and empirical investigation. Future research should examine whether the predictive power of DSF varies across different market conditions, monetary policy regimes, or credit cycles. Additionally, investigating the interaction between DSF and corporate investment decisions, as well as its relationship with credit market conditions, could provide deeper insights into the economic channels driving the return predictability. Researchers might also explore whether combining DSF with machine learning techniques or alternative data sources could enhance its predictive power. Finally, examining the signal’s performance during extreme market events and its contribution to downside risk protection would provide valuable insights for risk management applications.



**Figure 1:** Times series of DSF percentiles and coverage.  
This figure plots descriptive statistics for DSF. Panel A shows cross-sectional percentiles of DSF over the sample. Panel B plots the monthly coverage of DSF relative to the universe of CRSP stocks with available market capitalizations.

**Table 1:** Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on DSF. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, [Fama and French \(1993\)](#) three-factor model, [Fama and French \(1993\)](#) three-factor model augmented with the [Carhart \(1997\)](#) momentum factor, [Fama and French \(2015\)](#) five-factor model, and the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor following [Fama and French \(2018\)](#). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the [Fama and French \(2015\)](#) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 198806 to 202406.

| Panel A: Excess returns and alphas on DSF-sorted portfolios                                       |                  |                  |                  |                  |                  |                  |
|---------------------------------------------------------------------------------------------------|------------------|------------------|------------------|------------------|------------------|------------------|
|                                                                                                   | (L)              | (2)              | (3)              | (4)              | (H)              | (H-L)            |
| $r^e$                                                                                             | 0.62<br>[2.78]   | 0.79<br>[3.79]   | 0.70<br>[3.57]   | 0.72<br>[3.54]   | 0.86<br>[3.95]   | 0.23<br>[2.71]   |
| $\alpha_{CAPM}$                                                                                   | -0.14<br>[-2.45] | 0.09<br>[1.35]   | 0.06<br>[0.78]   | 0.03<br>[0.52]   | 0.11<br>[2.04]   | 0.26<br>[2.93]   |
| $\alpha_{FF3}$                                                                                    | -0.13<br>[-2.24] | 0.06<br>[0.96]   | 0.01<br>[0.20]   | 0.01<br>[0.12]   | 0.14<br>[2.73]   | 0.27<br>[3.12]   |
| $\alpha_{FF4}$                                                                                    | -0.09<br>[-1.52] | 0.06<br>[0.91]   | 0.02<br>[0.26]   | 0.02<br>[0.26]   | 0.13<br>[2.44]   | 0.21<br>[2.47]   |
| $\alpha_{FF5}$                                                                                    | -0.11<br>[-1.90] | -0.04<br>[-0.59] | -0.15<br>[-2.36] | -0.10<br>[-1.65] | 0.17<br>[3.25]   | 0.28<br>[3.19]   |
| $\alpha_{FF6}$                                                                                    | -0.08<br>[-1.32] | -0.03<br>[-0.51] | -0.14<br>[-2.14] | -0.08<br>[-1.41] | 0.16<br>[2.96]   | 0.23<br>[2.65]   |
| Panel B: <a href="#">Fama and French (2018)</a> 6-factor model loadings for DSF-sorted portfolios |                  |                  |                  |                  |                  |                  |
| $\beta_{MKT}$                                                                                     | 1.01<br>[68.74]  | 0.99<br>[60.59]  | 0.94<br>[58.60]  | 0.99<br>[67.91]  | 1.00<br>[75.43]  | -0.02<br>[-0.78] |
| $\beta_{SMB}$                                                                                     | 0.02<br>[0.94]   | 0.00<br>[0.15]   | -0.00<br>[-0.07] | -0.03<br>[-1.39] | -0.05<br>[-2.64] | -0.07<br>[-2.21] |
| $\beta_{HML}$                                                                                     | -0.09<br>[-3.59] | 0.03<br>[0.95]   | 0.04<br>[1.50]   | -0.01<br>[-0.24] | -0.09<br>[-3.93] | 0.00<br>[0.05]   |
| $\beta_{RMW}$                                                                                     | -0.05<br>[-1.94] | 0.13<br>[4.49]   | 0.28<br>[9.47]   | 0.17<br>[6.39]   | -0.03<br>[-1.11] | 0.03<br>[0.64]   |
| $\beta_{CMA}$                                                                                     | 0.06<br>[1.67]   | 0.19<br>[4.61]   | 0.21<br>[5.04]   | 0.14<br>[3.74]   | -0.10<br>[-2.92] | -0.16<br>[-2.87] |
| $\beta_{UMD}$                                                                                     | -0.06<br>[-4.26] | -0.01<br>[-0.57] | -0.02<br>[-1.46] | -0.02<br>[-1.59] | 0.02<br>[1.89]   | 0.08<br>[3.99]   |
| Panel C: Average number of firms ( $n$ ) and market capitalization ( $me$ )                       |                  |                  |                  |                  |                  |                  |
| $n$                                                                                               | 832              | 563              | 522              | 570              | 784              |                  |
| $me$ (\$10 <sup>6</sup> )                                                                         | 2927             | 2434             | 2381             | 2893             | 4206             |                  |



**Table 2:** Robustness to sorting methodology & trading costs

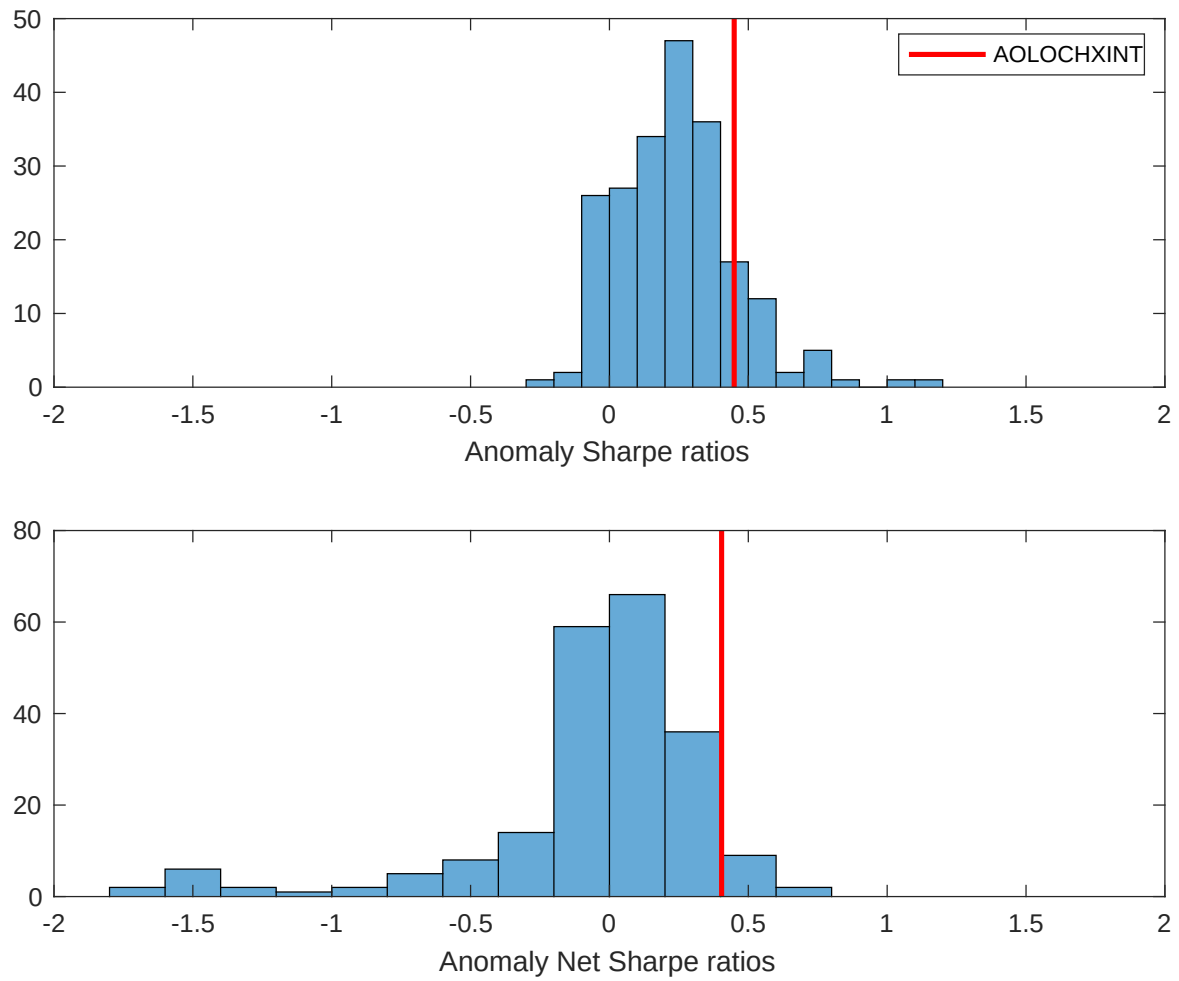
This table evaluates the robustness of the choices made in the DSF strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 198806 to 202406.

| Panel A: Gross Returns and Alphas                                        |        |         |                    |                          |                         |                         |                         |                         |
|--------------------------------------------------------------------------|--------|---------|--------------------|--------------------------|-------------------------|-------------------------|-------------------------|-------------------------|
| Portfolios                                                               | Breaks | Weights | $r^e$              | $\alpha_{\text{CAPM}}$   | $\alpha_{\text{FF3}}$   | $\alpha_{\text{FF4}}$   | $\alpha_{\text{FF5}}$   | $\alpha_{\text{FF6}}$   |
| Quintile                                                                 | NYSE   | VW      | 0.23<br>[2.71]     | 0.26<br>[2.93]           | 0.27<br>[3.12]          | 0.21<br>[2.47]          | 0.28<br>[3.19]          | 0.23<br>[2.65]          |
| Quintile                                                                 | NYSE   | EW      | 0.19<br>[3.05]     | 0.21<br>[3.29]           | 0.21<br>[3.36]          | 0.14<br>[2.29]          | 0.20<br>[3.05]          | 0.14<br>[2.21]          |
| Quintile                                                                 | Name   | VW      | 0.27<br>[2.59]     | 0.29<br>[2.77]           | 0.30<br>[2.94]          | 0.21<br>[2.10]          | 0.27<br>[2.56]          | 0.20<br>[1.90]          |
| Quintile                                                                 | Cap    | VW      | 0.28<br>[2.87]     | 0.29<br>[2.92]           | 0.31<br>[3.19]          | 0.25<br>[2.53]          | 0.36<br>[3.54]          | 0.30<br>[2.97]          |
| Decile                                                                   | NYSE   | VW      | 0.40<br>[3.03]     | 0.41<br>[3.08]           | 0.44<br>[3.31]          | 0.30<br>[2.32]          | 0.44<br>[3.18]          | 0.32<br>[2.39]          |
| Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas |        |         |                    |                          |                         |                         |                         |                         |
| Portfolios                                                               | Breaks | Weights | $r_{\text{net}}^e$ | $\alpha_{\text{CAPM}}^*$ | $\alpha_{\text{FF3}}^*$ | $\alpha_{\text{FF4}}^*$ | $\alpha_{\text{FF5}}^*$ | $\alpha_{\text{FF6}}^*$ |
| Quintile                                                                 | NYSE   | VW      | 0.21<br>[2.44]     | 0.25<br>[2.90]           | 0.26<br>[3.07]          | 0.23<br>[2.69]          | 0.28<br>[3.17]          | 0.24<br>[2.84]          |
| Quintile                                                                 | NYSE   | EW      | -0.02<br>[-0.23]   | 0.01<br>[0.10]           |                         |                         |                         |                         |
| Quintile                                                                 | Name   | VW      | 0.24<br>[2.34]     | 0.29<br>[2.81]           | 0.30<br>[2.96]          | 0.25<br>[2.47]          | 0.28<br>[2.71]          | 0.23<br>[2.29]          |
| Quintile                                                                 | Cap    | VW      | 0.26<br>[2.64]     | 0.29<br>[2.98]           | 0.31<br>[3.22]          | 0.27<br>[2.83]          | 0.35<br>[3.58]          | 0.31<br>[3.23]          |
| Decile                                                                   | NYSE   | VW      | 0.37<br>[2.80]     | 0.38<br>[2.83]           | 0.40<br>[3.03]          | 0.32<br>[2.45]          | 0.40<br>[2.96]          | 0.33<br>[2.49]          |

**Table 3:** Conditional sort on size and DSF

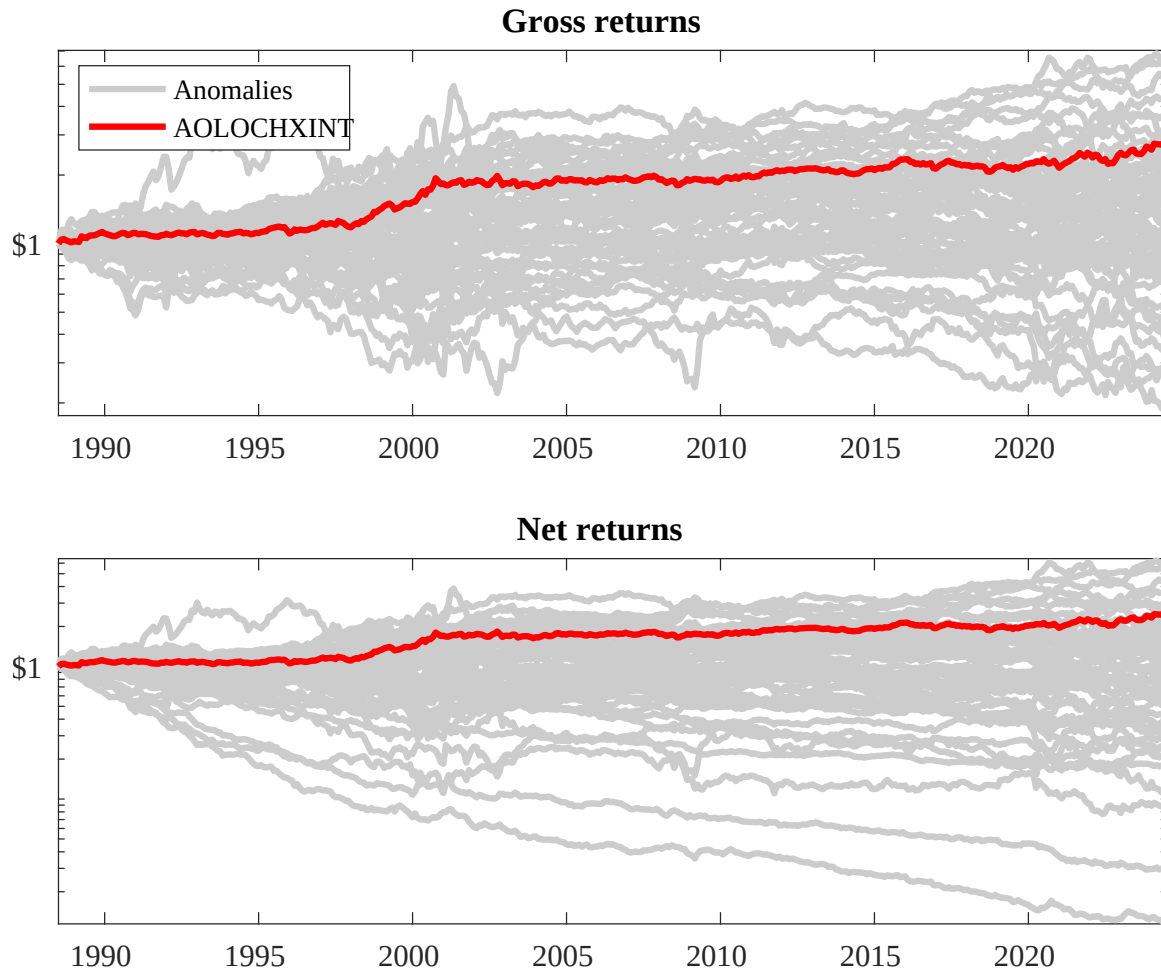
This table presents results for conditional double sorts on size and DSF. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on DSF. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high DSF and short stocks with low DSF .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 198806 to 202406.

| Panel A: portfolio average returns and time-series regression results |               |                |                |                |                |                |                                                    |                 |                |                |                |                |
|-----------------------------------------------------------------------|---------------|----------------|----------------|----------------|----------------|----------------|----------------------------------------------------|-----------------|----------------|----------------|----------------|----------------|
| Size quintiles                                                        | DSF Quintiles |                |                |                |                | DSF Strategies |                                                    |                 |                |                |                |                |
|                                                                       |               | (L)            | (2)            | (3)            | (4)            | (H)            | $r^e$                                              | $\alpha_{CAPM}$ | $\alpha_{FF3}$ | $\alpha_{FF4}$ | $\alpha_{FF5}$ | $\alpha_{FF6}$ |
|                                                                       | (1)           | 0.55<br>[1.64] | 0.80<br>[2.43] | 0.72<br>[1.99] | 0.82<br>[2.49] | 0.75<br>[2.25] | 0.20<br>[2.04]                                     | 0.18<br>[1.78]  | 0.16<br>[1.62] | 0.14<br>[1.37] | 0.19<br>[1.80] | 0.17<br>[1.58] |
|                                                                       | (2)           | 0.70<br>[2.26] | 0.74<br>[2.49] | 0.86<br>[2.95] | 0.75<br>[2.64] | 0.74<br>[2.43] | 0.04<br>[0.39]                                     | 0.09<br>[0.77]  | 0.11<br>[1.03] | 0.06<br>[0.53] | 0.11<br>[0.96] | 0.06<br>[0.54] |
|                                                                       | (3)           | 0.74<br>[2.72] | 0.72<br>[2.70] | 0.81<br>[3.02] | 0.83<br>[3.20] | 0.81<br>[2.93] | 0.07<br>[0.64]                                     | 0.06<br>[0.52]  | 0.09<br>[0.81] | 0.02<br>[0.21] | 0.19<br>[1.75] | 0.13<br>[1.21] |
|                                                                       | (4)           | 0.82<br>[3.09] | 0.72<br>[3.01] | 0.70<br>[2.97] | 0.83<br>[3.54] | 0.90<br>[3.36] | 0.08<br>[0.84]                                     | 0.07<br>[0.73]  | 0.10<br>[1.13] | 0.05<br>[0.55] | 0.17<br>[1.81] | 0.12<br>[1.28] |
|                                                                       | (5)           | 0.61<br>[2.76] | 0.81<br>[4.08] | 0.69<br>[3.53] | 0.67<br>[3.31] | 0.97<br>[4.39] | 0.37<br>[3.36]                                     | 0.37<br>[3.32]  | 0.39<br>[3.54] | 0.31<br>[2.85] | 0.42<br>[3.74] | 0.35<br>[3.15] |
| Panel B: Portfolio average number of firms and market capitalization  |               |                |                |                |                |                |                                                    |                 |                |                |                |                |
| Size quintiles                                                        | DSF Quintiles |                |                |                |                | DSF Quintiles  |                                                    |                 |                |                |                |                |
|                                                                       |               | Average $n$    |                |                |                |                | Average market capitalization (\$10 <sup>6</sup> ) |                 |                |                |                |                |
|                                                                       |               | (L)            | (2)            | (3)            | (4)            | (H)            | (L)                                                | (2)             | (3)            | (4)            | (H)            |                |
|                                                                       | (1)           | 344            | 343            | 340            | 341            | 343            | 36                                                 | 38              | 38             | 37             | 38             |                |
|                                                                       | (2)           | 108            | 108            | 108            | 108            | 108            | 72                                                 | 74              | 74             | 74             | 74             |                |
|                                                                       | (3)           | 76             | 77             | 76             | 77             | 77             | 131                                                | 133             | 134            | 134            | 133            |                |
|                                                                       | (4)           | 67             | 66             | 67             | 66             | 67             | 295                                                | 303             | 300            | 308            | 304            |                |
| (5)                                                                   | 61            | 61             | 61             | 61             | 61             | 2517           | 2051                                               | 2131            | 2508           | 2902           |                |                |



**Figure 2:** Distribution of Sharpe ratios.

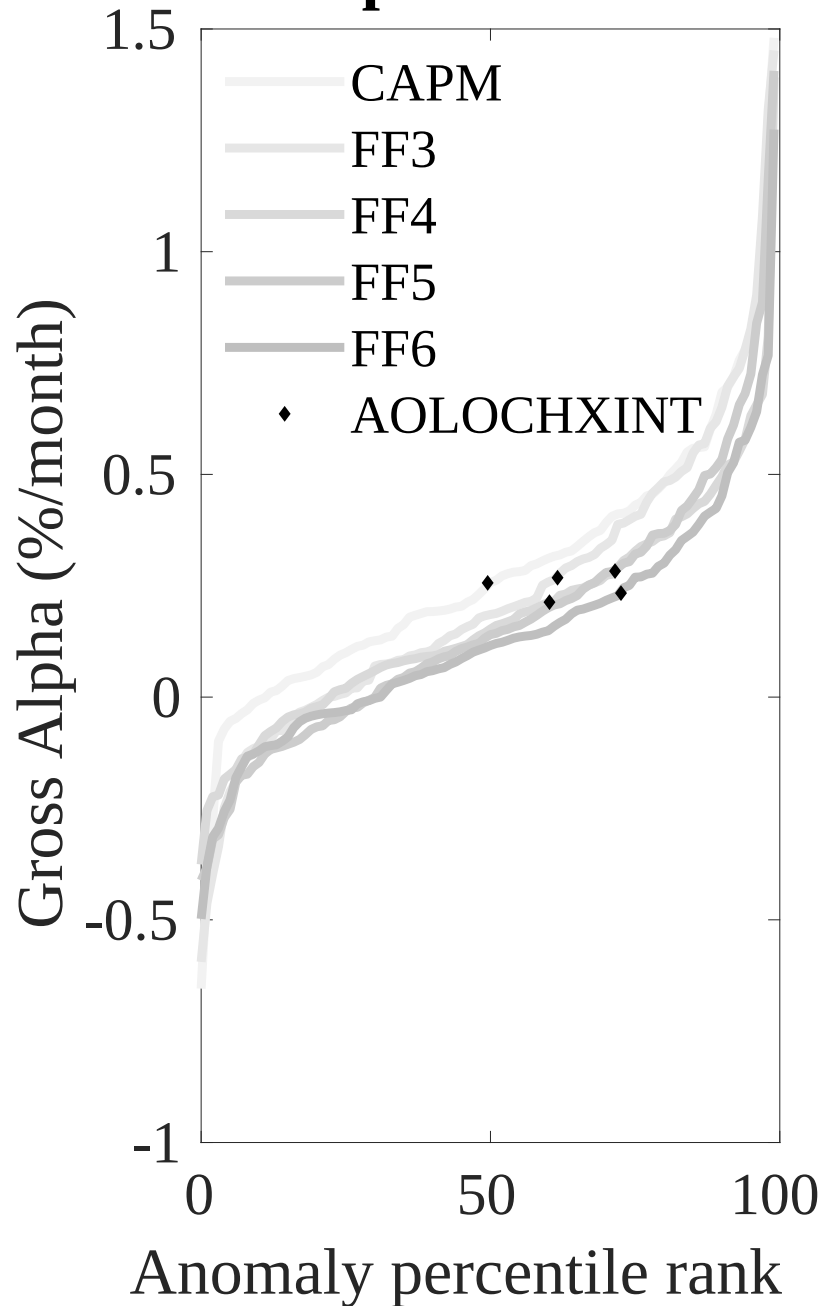
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the DSF with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.



**Figure 3:** Dollar invested.

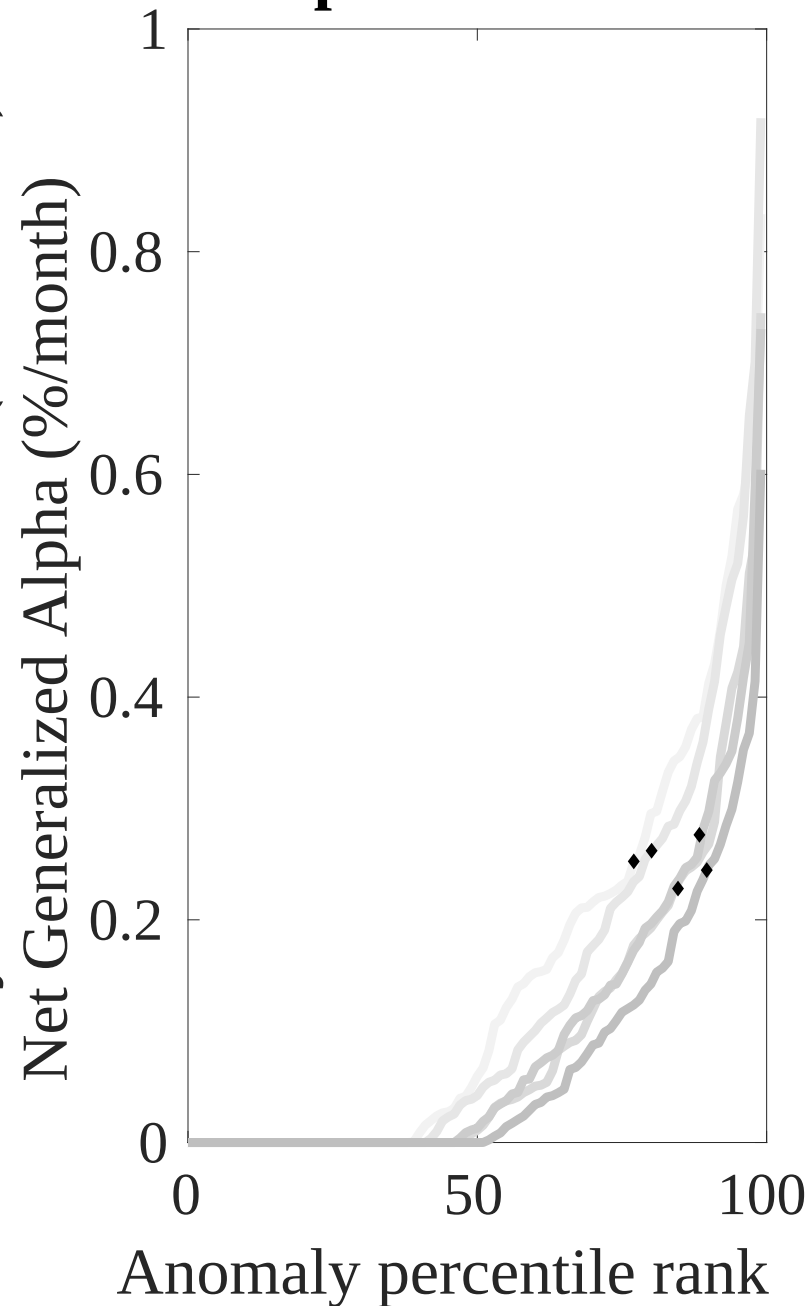
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the DSF trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

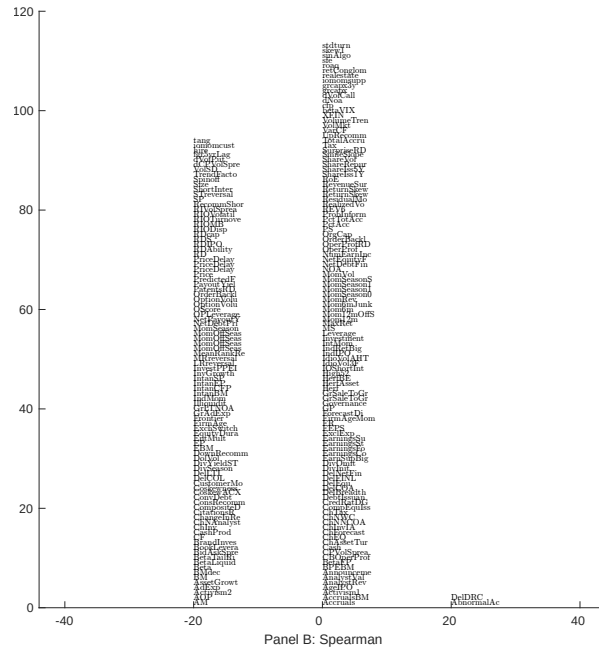
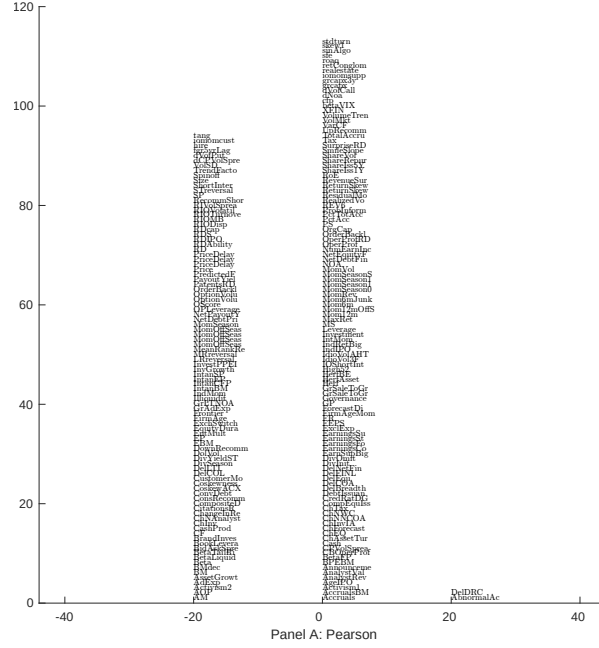
## Gross Alpha Percentiles



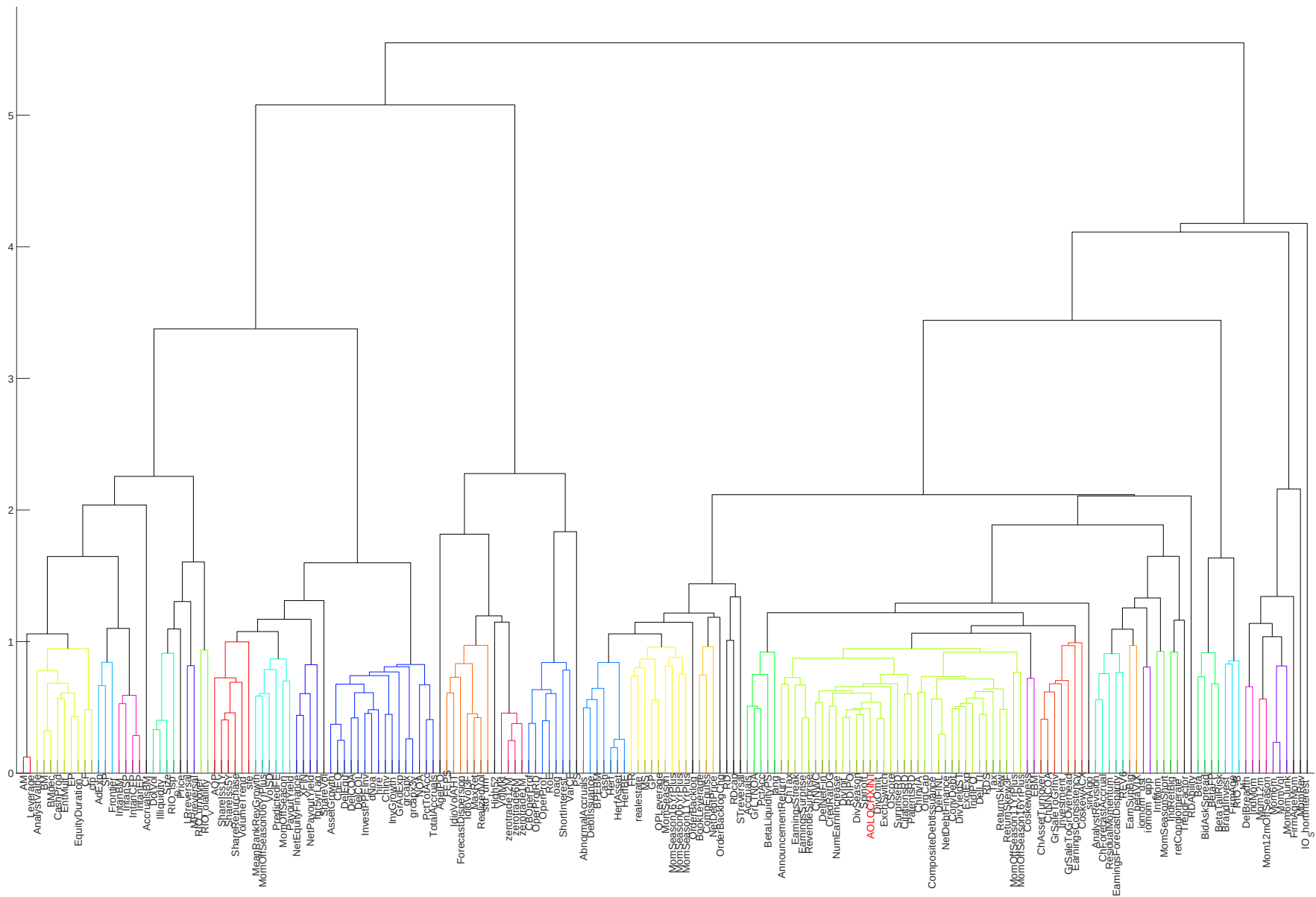
Novy-Marx and Velikov (RFS, 2016)

## Net Alpha Percentiles



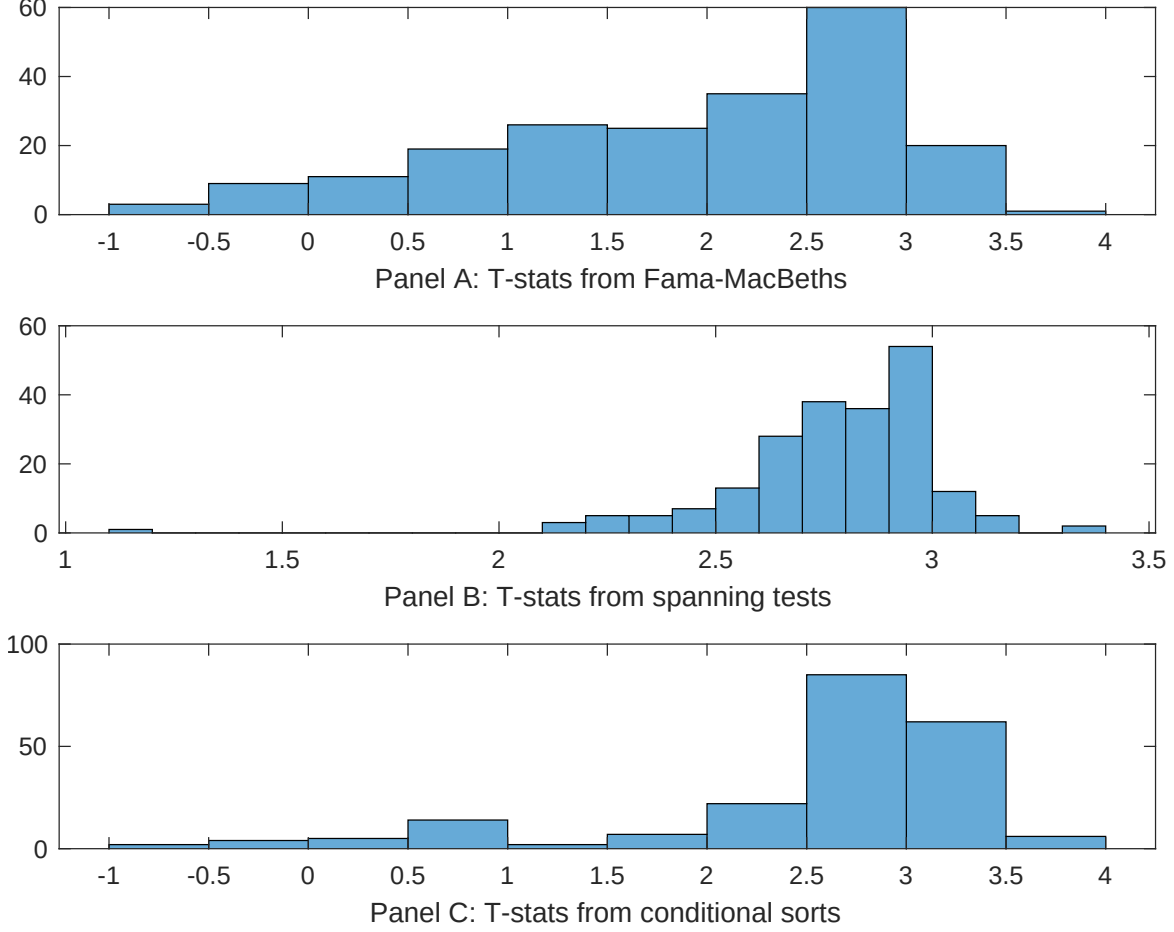


**Figure 5:** Distribution of correlations. This figure plots a name histogram of correlations of 209 filtered anomaly signals with DSF. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.



**Figure 6:** Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.



**Figure 7:** Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of DSF conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on  $\beta_{DSF}$  from Fama-MacBeth regressions of the form  $r_{i,t} = \alpha + \beta_{DSF}DSF_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$ , where  $X$  stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on  $\alpha$  from spanning tests of the form:  $r_{DSF,t} = \alpha + \beta r_{X,t} + \epsilon_t$ , where  $r_{X,t}$  stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on DSF. Stocks are finally grouped into five DSF portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted DSF trading strategies conditioned on each of the 209 filtered anomalies.



**Table 4:** Fama-MacBeths controlling for most closely related anomalies

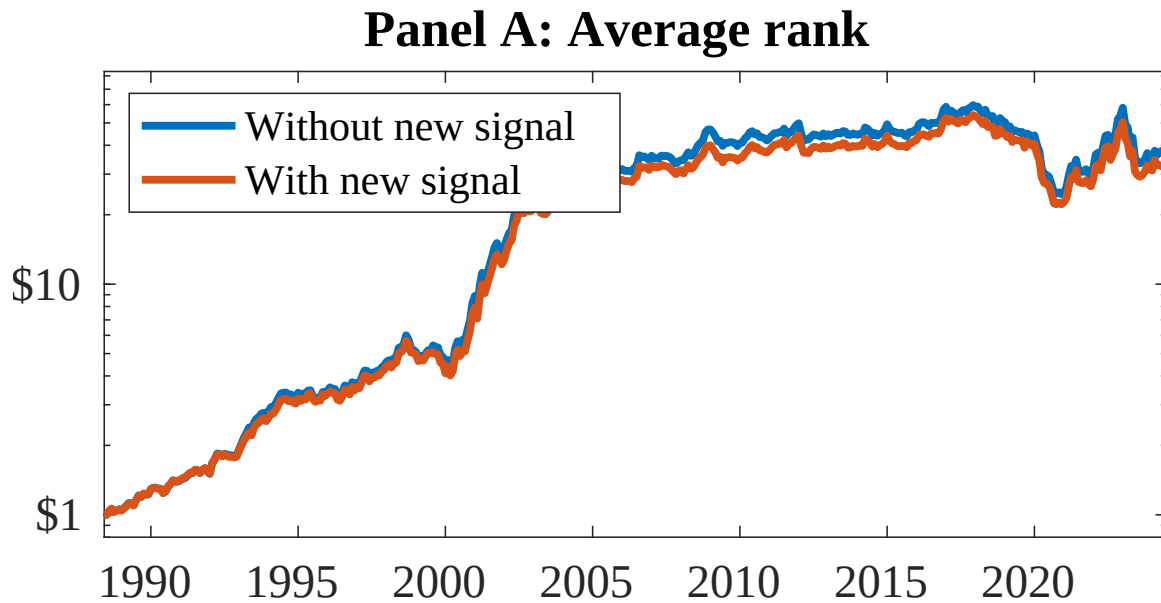
This table presents Fama-MacBeth results of returns on DSF. and the six most closely related anomalies. The regressions take the following form:  $r_{i,t} = \alpha + \beta_{DSF}DSF_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$ . The six most closely related anomalies,  $X$ , are Change in current operating liabilities, Book-to-market and accruals, Efficient frontier index, Change in Taxes, Price, Total accruals. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the  $R^2$  from the spanning tests in Figure 7, Panel B. The sample period is 198806 to 202406.

|                 |                |                |                |                |                |                |                  |
|-----------------|----------------|----------------|----------------|----------------|----------------|----------------|------------------|
| Intercept       | 0.11<br>[3.95] | 0.66<br>[1.85] | 0.11<br>[3.87] | 0.11<br>[3.90] | 0.11<br>[2.97] | 0.11<br>[3.86] | 0.24<br>[3.38]   |
| DSF             | 0.35<br>[3.05] | 0.80<br>[1.59] | 0.25<br>[1.68] | 0.31<br>[2.71] | 0.36<br>[3.21] | 0.26<br>[2.21] | 0.28<br>[0.42]   |
| Anomaly 1       | 0.19<br>[4.46] |                |                |                |                |                | 1.00<br>[0.58]   |
| Anomaly 2       |                | 0.14<br>[4.58] |                |                |                |                | -0.25<br>[-3.26] |
| Anomaly 3       |                |                | 0.44<br>[4.46] |                |                |                | 0.20<br>[6.41]   |
| Anomaly 4       |                |                |                | 0.14<br>[5.38] |                |                | 0.20<br>[2.25]   |
| Anomaly 5       |                |                |                |                | 0.45<br>[0.88] |                | -0.75<br>[-0.81] |
| Anomaly 6       |                |                |                |                |                | 0.47<br>[1.64] | 0.67<br>[1.05]   |
| # months        | 432            | 415            | 427            | 432            | 432            | 432            | 267              |
| $\bar{R}^2(\%)$ | 0              | 1              | 1              | 0              | 1              | 0              | 0                |

**Table 5:** Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the DSF trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form:  $r_t^{DSF} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$ , where  $X_k$  indicates each of the six most-closely related anomalies and  $f_j$  indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies,  $X$ , are Change in current operating liabilities, Book-to-market and accruals, Efficient frontier index, Change in Taxes, Price, Total accruals. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the  $R^2$  from the spanning tests in Figure 7, Panel B. The sample period is 198806 to 202406.

|                 |                   |                   |                   |                   |                   |                   |                  |
|-----------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|------------------|
| Intercept       | 0.22<br>[2.54]    | 0.26<br>[2.99]    | 0.26<br>[2.93]    | 0.26<br>[3.05]    | 0.26<br>[2.96]    | 0.26<br>[3.03]    | 0.24<br>[2.71]   |
| Anomaly 1       | -12.60<br>[-3.09] |                   |                   |                   |                   |                   | -8.98<br>[-2.02] |
| Anomaly 2       |                   | -1.21<br>[-0.96]  |                   |                   |                   |                   | -1.07<br>[-0.85] |
| Anomaly 3       |                   |                   | -0.40<br>[-0.15]  |                   |                   |                   | 0.77<br>[0.27]   |
| Anomaly 4       |                   |                   |                   | 17.50<br>[5.03]   |                   |                   | 15.42<br>[4.24]  |
| Anomaly 5       |                   |                   |                   |                   | 0.34<br>[0.10]    |                   | -0.42<br>[-0.12] |
| Anomaly 6       |                   |                   |                   |                   |                   | -7.57<br>[-1.87]  | -1.45<br>[-0.33] |
| mkt             | -1.62<br>[-0.75]  | -1.95<br>[-0.89]  | -2.03<br>[-0.92]  | -3.61<br>[-1.68]  | -1.94<br>[-0.86]  | -1.74<br>[-0.80]  | -3.17<br>[-1.41] |
| smb             | -7.48<br>[-2.39]  | -5.70<br>[-1.78]  | -6.07<br>[-1.79]  | -6.38<br>[-2.09]  | -6.63<br>[-1.51]  | -6.89<br>[-2.19]  | -6.80<br>[-1.59] |
| hml             | 7.07<br>[1.65]    | 2.40<br>[0.58]    | 1.19<br>[0.30]    | 3.45<br>[0.92]    | 0.86<br>[0.22]    | 0.17<br>[0.05]    | 8.39<br>[1.76]   |
| rmw             | 2.44<br>[0.62]    | 1.36<br>[0.34]    | 1.90<br>[0.48]    | 1.98<br>[0.51]    | 2.33<br>[0.52]    | 0.52<br>[0.13]    | 1.05<br>[0.23]   |
| cma             | -10.72<br>[-1.81] | -16.70<br>[-2.97] | -17.09<br>[-3.03] | -15.25<br>[-2.81] | -17.38<br>[-3.10] | -12.10<br>[-1.95] | -9.37<br>[-1.48] |
| umd             | 7.23<br>[3.74]    | 7.10<br>[3.40]    | 7.55<br>[2.86]    | 3.34<br>[1.60]    | 8.07<br>[2.73]    | 7.20<br>[3.66]    | 2.89<br>[0.91]   |
| # months        | 432               | 428               | 428               | 432               | 432               | 432               | 428              |
| $\bar{R}^2(\%)$ | 10                | 8                 | 8                 | 13                | 8                 | 9                 | 13               |



**Figure 8:** Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 158 anomalies. The red solid lines indicate combination trading strategies that utilize the 158 anomalies as well as DSF. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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